

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou)Sep 2023 - Jun 2027 (expected)

BSc in Data Science and Big Data Technology · Dean's List x3

- CGA 3.807/4.3; Major CGA 3.837/4.3.
- Selected coursework: Advanced Algorithms and Data Structures (A+), Design and Analysis of Algorithms (A), Database Management Systems (A+), Linear Algebra (A+), Honors Calculus I/II (A+/A+), Applied Statistics (A-), Machine Learning (A-), Cloud Computing and Big Data Systems (A-).

Osaka UniversityFall 2025

Exchange Study · School of Engineering Science

- Mathematical Science for Engineering Science (A), Quantum Physics II (A), Fundamentals of Quantum Electronics and Quantum Information (A), Circuit Theory III (S, highest grade).

INTERNSHIP EXPERIENCE

Xuanyuan Private Fund Management (Guangdong) Co., Ltd.Jul 2026 - Present

Quantitative Investment Modeling Research Intern · Guangzhou

- Market microstructure research:** build fixed-volume bars from A-share Level-2 trades and study execution topology, active/passive order structure, direction continuity, price paths and duration across 2025H1 market-wide data.
- Event response and evaluation:** construct post-confirmation quote-response analysis at 5/30/60/180/300-second horizons; enforce point-in-time availability, date-block confidence intervals and temporal holdouts to separate mechanism evidence from tradable-signal claims.
- Research findings:** distinguish one-sided penetration, dense two-sided exchange and sparse long-gap execution paths; retain low-degree-of-freedom mechanism states such as direction support × path efficiency without claiming a stable minute alpha before incremental validation.
- Data and training engineering:** maintain multi-year, market-wide minute research pipelines with memory-mapped storage, parallel I/O, temporal/cross-asset alignment and batch feature/label generation; redesigned transposed cross-day DataLoader access for about 4× faster data loading.

RESEARCH EXPERIENCE

Final Year Project - Reliable Multi-Agent Financial ForecastingJul 2026 - Present

Undergraduate Research Project · Supervisors: Prof. Hao Xue / Prof. Yuxuan Liang

- Study reliable forecasting and decision-making under noisy, changing financial information, focusing on evidence aggregation, claim-level verification, calibration, point-in-time provenance and temporal grounding.
- Current stage: framework and evaluation design, model/API and compute preparation, and automated experiment-environment setup; no premature claim of completed experimental results.

LENS: Large Pre-trained Transformer for Exploring Financial Time Series RegularitiesJun 2024 - Oct 2024

Undergraduate Research Assistant / Co-author · Prof. Guang Zhang

- Built terabyte-scale financial time-series preprocessing and research infrastructure with chunked loading, parallel cleaning, statistical aggregation, sample organization and model-input construction.
- Internal benchmarks achieved up to 13× faster data cleaning and 6× faster statistical aggregation; work contributed to the ACM ICAIF 2025 paper (pp. 771-778).

Research-Activity Measurement and ForecastingSep 2026 - Present

Research Participant · Prof. Guang Zhang

- Contribute to an LLM-assisted project on measuring and forecasting scientific research activity; current work covers data collection, cleaning and early problem formulation, with emphasis on measurement targets, timestamp conventions and interpretability.

SpatialReflect: Spatially Disentangled Reflection for Controllable GenerationJun 2025 - May 2026

First-author Independent Research · HKUST(GZ)

- Independently completed problem formulation, training-free method design, theoretical analysis, implementation, controlled/ablation experiments and manuscript writing for multi-region diffusion generation; went through a full peer-review cycle.

PROJECT EXPERIENCE

ASC World University Supercomputer Competition - AI/HPC Optimization

2025 - 2026

Core Team Member · HKUST(GZ) representative team

- Optimized C++/CUDA inference workloads on NVIDIA A800 clusters using TensorRT, low-precision inference, layer/operator fusion, dynamic batching, GPU profiling/tuning, MPI and Slurm; handled CPU affinity, NUMA/cache locality and reproducible cluster experiments.
- For the UniFOLM-WMA inference task, reduced internal-test runtime from about 350s to 48s (~7.3×) and GPU memory from about 12.4GB to 4.8GB. Team won ASC 2026 First Prize and ASC 2025 Second Prize.

Deterministic L2 Order-Book Replay and Cross-Language Validation

2026

Personal systems project · C++20 / Rust / Python

- Implemented deterministic event replay ordered by information availability and an L2 price-level order-book consumer; used an independent Python oracle for cross-language consistency checks across 144 runs and 11,638 lines of book-state output.
- Benchmarked alternative merge/replay implementations and added sanitizer/conformance CI, emphasizing state modeling, correctness baselines, data-structure choices and benchmark-driven optimization.

SPY One-Day Conditional VaR Forecasting

Aug 2026

Independent financial time-series research project

- Built a leakage-safe rolling out-of-sample framework for 1% / 5% / 10% one-day-ahead conditional VaR; compared Historical Simulation, Student-t GARCH/GJR-GARCH, realized-measure quantile regression, MLP and GRU models.
- Froze development choices before the final test and evaluated with quantile loss, coverage and independence tests; GJR-GARCH was more stable across the three tail quantiles while small neural models showed no consistent advantage.

TECHNICAL SKILLS

C++: long-term user; strong command of C++17/20 and earlier modern C++ features, partial C++23 knowledge, and ongoing tracking of C++26. Comfortable with RAII, move semantics, templates/concepts, STL/ranges, lambdas and higher-order algorithms, coroutines, concurrency/multithreading, CMake, sanitizers, benchmarking and profiling; able to develop performance-sensitive data structures and systems components independently.

Rust: practical systems-development experience using ownership/borrowing, enum/match, traits/generics, Result/Option and Cargo; applied to reliable state components, event replay and state-isolation style problems.

Python / Data / ML: Python, NumPy, Polars/pandas, SciPy, scikit-learn, PyTorch, LightGBM, XGBoost, SQL; large arrays, memory mapping, parallel I/O, batch data pipelines and reproducible experimentation.

Quantitative Research: A-share Level-2/minute data, market microstructure, order lifecycle, factor/neural-feature research, TS/CS IC and RankIC, grouped returns, walk-forward/OOT validation, stability/decay/redundancy analysis, leakage and sample-definition audits.

Systems / HPC: Linux, Git, Docker, CUDA, TensorRT, MPI, Slurm; GPU/CPU profiling, CPU affinity, NUMA/cache locality, concurrency and reproducible cluster experiments.

Languages: Chinese (native); English (full-medium academic instruction and technical communication); Japanese (JLPT N1).

HONORS

ASC World University Supercomputer Competition 2026	First Prize (team)
ASC World University Supercomputer Competition 2025	Second Prize (team)
Chinese Physics Olympiad (CPhO)	Provincial First Prize ×2
HKUST(GZ) TradeMaster Cup 2026	5th Place (team leader)
MCM/ICM Mathematical Contest in Modeling 2025	Honorable Mention
INFH Lizhi Innovation Talent Scholarship 2025 / HKUST(GZ) Admission Scholarship 2023	Scholarships